

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** VALVES *****			821	TO BE USED FOR LINES SUBJECT TO AIV OR WHERE REQUIRED BY STRESS.
20							***** MODULAR VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
30										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
30	U04	VAV	MODULAR DBB	K03AC02	0.75	3	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
40	U04		MODULAR DBB	M03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
50	U04	VAW	MODULAR DBB	M03EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	3	ND9	USE ONLY WITH CONSENT OF MATERIALS ENGINEER.
60	U04	ND9	MODULAR DBB	M06AC02	1.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	NT5	TO BE USED IN NON-PIGGED LINES ONLY
70		VB2					***** BALL VALVES *****			VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
80	U04		BALL	A03AC01	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
90	U04		BALL	A03AC02	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
100	U04		BALL	A03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
110	U04		BALL	A03AC08	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
120	U04	VAV	BALL	B03AC01	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
130	U04	VAV	BALL	B03AC02	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
140	U04	VA5	BALL	B03AC03	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0		
150	U04	VA5	BALL	B03AC04	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0		
160	U04	VA5	BALL	B03AC07	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
170	U04	VA5	BALL	B03AC08	6	24	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
180	U04	VA5	BALL	B03JC07	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO B16.47 SERIES A	0	0		
190	U04	VA5	BALL	B03JC08	28	28	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0		
200							***** RISING STEM BALL VALVES *****				
210		VA5	BALL	R03AC06	0.75	8	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
220		VA5	BALL	R03AC08	6	10	BALL RISING STEM 300# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
230							***** BUTTERFLY VALVES *****				
240			BUTTERFLY	T03AC61	10	24	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
250			BUTTERFLY	T03JC61	28	28	BUTTERFLY 300# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A	0	0		
260							***** GATE VALVES *****				
270											
270	U04		GATE	W03AC70	3	8	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
280			GATE	W03AC71	8	24	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
290			GATE	W03JC71	28	28	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO B16.47 SERIES A	0	0		
300							***** GLOBE VALVES *****				
310											
310	U04		GLOBE	L03AC48	0.75	1.5	GLOBE 300# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
320			GLOBE	L03AC41	2	6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
330			GLOBE	L03AC42	8	10	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
340							***** CHECK VALVES *****				
350											
350	U04		CHECK	C03AC50	0.75	1.5	CHECK PISTON 300# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
360			CHECK	C03AC45	2	10	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
370			CHECK	C03AC52	12	24	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16	0	0		
380			CHECK	C03JC52	28	28	CHECK DUAL DBL FLG 300# RF LTCS A352 LCC TRIM 16 B16.47 SERIES A	0	0		
390							***** PIPE ABOVE GROUND *****				
400											
400			PIPE	5358512	0.75	0.75	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
410	U04			5358511	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
420			PIPE	5358508	1	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0		
430			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
440			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
450											
450			PIPE	9001257	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0		
460			PIPE	9001254	6	10	LINEPIPE SMLS WT= 6.35MM API-5L X60 BE	0	0		
470			PIPE	9001256	12	12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE	0	0		
480			PIPE	9001311	14	14	LINEPIPE SMLS WT=7.92MM API-5L X60 BE	0	0		
490			PIPE	9001485	16	18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE	0	0		
500			PIPE	9001493	20	20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE	0	0		
510			PIPE	9001488	24	24	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 95 °C	12/16/2024	BY OZAC	U04	Page 1 of 6	300K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
520											
520			PIPE	5358513	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
530											
530	U04	821	PIPE	9005244	16	16	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE	0	0		
540	U04	821	PIPE	9005241	20	20	LINEPIPE SMLS WT= 15.09 MM API-5L X60 BE	0	0		
550											
550							***** PIPE BELOW GROUND *****				
560											
560			PIPE	9001257CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0		
570			PIPE	9001254CTD	6	10	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0		
580			PIPE	9001256CTD	12	12	LINEPIPE SMLS WT= 7.14 MM API-5L X60 BE COATED	0	0		
590			PIPE	9001311CTD	14	14	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0		
600			PIPE	9001485CTD	16	18	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0		
610			PIPE	9001493CTD	20	20	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0		
620			PIPE	9001488CTD	24	24	LINEPIPE SMLS WT = 11.13 MM API-5L X60 BE COATED	0	0		
630			PIPE	9001490CTD	28	28	LINEPIPE SMLS WT = 11.91 MM API-5L X60 BE COATED	0	0		
640											
640							***** FITTINGS ABOVE & BELOW GROUND*****				
650											
650			ELBOW	5392816	0.75	0.75	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0		
660			ELBOW	5392600	1	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
670			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0		
680			ELBOW	9006566	4	4	ELL BW 90 DEG LR WT 6.02 MM A860 WPHY 60	0	0		
690			ELBOW	9002190	6	10	ELL BW 90 DEG LR WT 6.35 MM A860 WPHY 60	0	0		
700			ELBOW	9006477	12	12	ELL BW 90 DEG LR WT 7.14 MM A860 WPHY 60	0	0		
710			ELBOW	9002157	16	18	ELL BW 90 DEG LR WT 8.74MM A860 WPHY 60	0	0		
720			ELBOW	9006472	20	20	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60-W	0	0		
730			ELBOW	9006575	24	24	ELL BW 90 DEG LR WT 11.13 MM A860 WPHY 60-W	0	0		
740			ELBOW	5392822	0.75	0.75	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0		
750			ELBOW	5392606	1	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
760			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0		
770			ELBOW	9006573	4	4	ELL BW 45 DEG LR WT 6.02MM A860 WPHY 60	0	0		
780	U04			9006571	6	6	ELL BW 45 DEG LR WT 6.35MM A860 WPHY 60	0	0		
790			ELBOW	9006593	16	18	ELL BW 45 DEG LR WT 8.74MM A860 WPHY 60	0	0		
800			ELBOW	9006570	20	20	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60-W	0	0		
810			ELBOW	9006577	24	24	ELL BW 45 DEG LR WT 11.13MM A860 WPHY 60-W	0	0		
820											
820			TEE	5392825	0.75	0.75	TEE BW SCH 160 LTCS A420 WPL6	0	0		
830			TEE	5392609	1	2	TEE BW SCH XS LTCS A420 WPL6	0	0		
840			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0		
850		NT5	TEE	9301610	4	4	TEE BW WT 6.02 MM LTCS A860 WPHY-60	0	0		
860	U04	NT5	TEE	9301609	6	10	TEE BW WT 6.35 MM LTCS A860 WPHY-60	0	0		
870		NT5	TEE	9301605	12	12	TEE BW WT 7.14 MM LTCS A860 WPHY-60	0	0		
880		NT5	TEE	9002113	16	18	TEE BW WT 8.7MM A860 WPHY 60	0	0		
890		NT5	TEE	9004210	20	20	TEE BW WT 9.53MM A860 WPHY 60-W	0	0		
900		NT5	TEE	9006638	24	24	TEE BW WT 11.13MM A860 WPHY 60-W	0	0		
910			TEE RED	5392469	0.75	0.75	TEE RED BW SCH XS/160 LTCS A420 WPL6	1	1.5		
920			TEE RED	5392661	1	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1.5	2		
930			TEE RED	5392454	1.5	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3		
940		NT5	TEE RED	9006664	2	2	TEE RED BW WT 6.02 MM / SCH XS LTCS A860 WPHY 60	4	4		
950		NT5	TEE RED	9006659	3	3	TEE RED BW WT 6.02 MM / STD LTCS A860 WPHY 60	4	4		
960		NT5	TEE RED	9006658	3	3	TEE RED BW WT 6.35 MM / STD LTCS A860 WPHY 60	6	6		
970		NT5	TEE RED	9006605	4	4	TEE RED BW WT 6.35 MM / 6.02 MM LTCS A860 WPHY 60	6	10		
980		NT5	TEE RED	9006014	6	6	TEE RED BW WT 6.35 MM LTCS A860 WPHY 60	8	10		
990			TEE RED	9004123	6	8	TEE RED BW WT 8.74 MM/6.35 MM LTCS A860 WPHY 60	16	18		
1000			TEE RED	9004149	8	10	TEE RED BW WT 9.53 MM/6.35 MM LTCS A860 WPHY 60-W	20	20		
1010		NT5	TEE RED	9006484	16	18	TEE RED BW WT 9.53 MM/8.74 MM LTCS A860 WPHY 60-W	20	20		
1020		NT5	TEE RED	9006652	10	10	TEE RED BW WT 11.13 MM/6.35 MM LTCS A860 WPHY 60-W	24	24		
1030		NT5	TEE RED	9006639	16	18	TEE RED BW WT 11.13 MM/8.74 MM LTCS A860 WPHY 60-W	24	24		
1040		NT5	TEE RED	9006640	20	20	TEE RED BW WT 11.13 MM/9.53 MM LTCS A860 WPHY 60-W	24	24		
1050	U04	821	TEE RED	9006685	12	12	TEE RED BW WT 15.09 MM Taper Bored to 9.53MM /11.13 MM LTCS A860 WPHY 60	20	20		
1060	U04	821	TEE RED	9006625	16	16	TEE RED BW WT 15.09 MM Taper Bored to 9.53MM /12.7 MM LTCS A860 WPHY 60	20	20		
1070	U04	821	TEE RED	9006671	16	16	TEE RED BW WT 17.48MM Taper Bored to 11.13MM /12.70MM LTCS A860 WPHY 60	24	24		
1080	U04	821	TEE RED	9006669	16	16	TEE RED BW WT 46.03MM Taper Bored to 11.13MM/30.95 MM LTCS A860 WPHY 60	24	24		
1090	U04	NT5	TEE RED	9006020	12	12	TEE RED BW WT 8.74 MM/7.14 MM LTCS A860 WPHY 60	16	16		
1100	U04	NT5	TEE RED	9006606	12	12	TEE RED BW WT 9.53 MM/7.14 MM LTCS A860 WPHY 60	20	20		
1110											
1110			CAP	5392826	0.75	0.75	CAP BW SCH 160 LTCS A420 WPL6	0	0		
1120			CAP	5392610	1	2	CAP BW SCH XS LTCS A420 WPL6	0	0		
1130			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0		
1140			CAP	9005062	4	4	CAP BW WT 6.02 MM LTCS A860 WPHY-60	0	0		
1150			CAP	9005055	6	10	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 95 °C	12/16/2024	BY OZAC	U04	Page 2 of 6	300K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1160			CAP	9005057	12	12	CAP BW WT 7.14 MM LTCS A860 WPHY-60	0	0	
1170			CAP	9005058	14	14	CAP BW WT 7.92 MM LTCS A860 WPHY-60	0	0	
1180			CAP	9005059	16	18	CAP BW WT 8.74 MM LTCS A860 WPHY-60	0	0	
1190			CAP	9005060	20	20	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0	
1200			CAP	9005064	24	24	CAP BW WT 11.13 MM LTCS A860 WPHY-60	0	0	
1210			CAP	9005065	28	28	CAP BW WT 11.91 MM LTCS A860 WPHY-60	0	0	
1220										
1220			RED CONC	5392462	0.75	0.75	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1	2	
1230			RED CONC	5392458	1	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1.5	2	
1240			RED ECC	5392465	0.75	0.75	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1	2	
1250			RED ECC	5392461	1	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1.5	2	
1260			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3	
1270			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3	
1280			RED CONC	9006918	1.5	3	REDUCER CONC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4	
1290			RED ECC	9001372	1.5	3	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4	
1300			RED CONC	9006915	3	3	REDUCER CONC WT 6.35 MM / XS LTCS A860 WPHY-60	6	6	
1310	U04		RED ECC	9002510	3	3	REDUCER ECC WT 6.35 MM/STD LTCS A860 WPHY-60	6	6	
1320			RED ECC	9002312	4	4	REDUCER ECC WT 6.35 MM / 6.02 MM LTCS A860 WPHY-60	6	10	
1330			RED ECC	9002266	6	8	REDUCER ECC WT 6.35 MM / 6.35 MM LTCS A860 WPHY-60	8	10	
1340			RED ECC	9002269	6	10	REDUCER ECC WT 7.14 MM / 6.35 MM LTCS A860 WPHY-60	12	12	
1350			RED ECC	9002272	6	10	REDUCER ECC WT 7.92 MM / 6.35 MM LTCS A860 WPHY-60	14	14	
1360			RED ECC	9002270	12	12	REDUCER ECC WT 7.92 MM / 7.14 MM LTCS A860 WPHY-60	14	14	
1370			RED ECC	9002292	8	10	REDUCER ECC WT 8.74 MM / 6.35 MM LTCS A860 WPHY-60	16	18	
1380			RED ECC	9002294	12	12	REDUCER ECC WT 8.74 MM / 7.14 LTCS A860 WPHY-60	16	18	
1390			RED ECC	9002293	14	14	REDUCER ECC WT 8.74 MM / 7.92 LTCS A860 WPHY-60	16	18	
1400			RED ECC	9002291	16	16	REDUCER ECC WT 8.74 MM LTCS A860 WPHY-60	18	18	
1410	U04	821	RED ECC	9006506	10	10	REDUCER ECC WT 12.70MM / 11.13 MM LTCS A860 WPHY-60	16	16	
1420	U04	821	RED ECC	9006504	10	10	REDUCER ECC WT 30.95MM / 21.44 MM LTCS A860 WPHY-60	16	16	
1430	U04		RED ECC	9002221	12	12	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60	20	20	
1440	U04		RED ECC	9002251	14	14	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60	20	20	
1450	U04		RED ECC	9002396	18	18	REDUCER ECC WT 9.53 MM / 8.74 MM LTCS A860 WPHY-60	20	20	
1460			RED ECC	9004134	12	12	REDUCER ECC WT 9.53 MM / 7.14 MM LTCS A860 WPHY-60-W	20	20	
1470			RED ECC	9004135	14	14	REDUCER ECC WT 9.53 MM / 7.92MM LTCS A860 WPHY-60-W	20	20	
1480		B1	RED ECC	9004136	16	18	REDUCER ECC WT 9.53 MM / 8.74 LTCS A860 WPHY-60-W	20	20	
1490		B1	RED ECC	9004161	8	18	REDUCER ECC WT 11.13 MM / 8.74 MM LTCS A860 WPHY-60-W	12	24	
1500			RED ECC	9004162	20	20	REDUCER ECC WT 11.13 MM / 9.53 MM LTCS A860 WPHY-60-W	24	24	
1510			RED ECC	9004146	18	18	REDUCER ECC WT 11.91MM / 8.74MM LTCS A860 WPHY-60-W	28	28	
1520			RED ECC	9004147	20	20	REDUCER ECC WT 11.91 MM / 9.53 LTCS A860 WPHY-60-W	28	28	
1530			RED ECC	9004172	24	24	REDUCER ECC WT 11.91MM / 11.13MM LTCS A860 WPHY-60-W	28	28	
1540							***** INDUCTION BENDS ABOVE GROUND*****			
1550										
1550			INDUC BEND	9001810	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1560			INDUC BEND	9001813	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1570			INDUC BEND	9001814	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1580			INDUC BEND	9001815	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1590			INDUC BEND	9001816	16	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1600			INDUC BEND	9001817	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1610			INDUC BEND	9001899	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1620			INDUC BEND	9001397	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1630			INDUC BEND	9001820	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1640			INDUC BEND	9001823	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1650			INDUC BEND	9001824	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1660			INDUC BEND	9001825	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1670			INDUC BEND	9001826	16	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1680			INDUC BEND	9001827	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1690	U04		INDUC BEND	9005326	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1700			INDUC BEND	9005326	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1710			INDUC BEND	9001500	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1720			INDUC BEND	9001830	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1730			INDUC BEND	9001833	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1740			INDUC BEND	9001834	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1750			INDUC BEND	9001835	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1760			INDUC BEND	9001836	16	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1770			INDUC BEND	9001837	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1780	U04		INDUC BEND	9005331	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1790			INDUC BEND	9005331	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1800			INDUC BEND	9002001	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
1810										
1810							***** INDUCTION BENDS BELOW GROUND*****			
1820										
1820			INDUC BEND	9001810CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	

	TENGINZCHEVROI TCO TENGINZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: - 40 / 95 °C	12/16/2024	BY OZAC	U04	Page 3 of 6	300K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1830			INDUC BEND	9001813CTD	6	10	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1840			INDUC BEND	9001814CTD	12	12	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1850			INDUC BEND	9001815CTD	14	14	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1860			INDUC BEND	9001816CTD	16	18	INDUCTION BEND 90 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1870			INDUC BEND	9001817CTD	20	20	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1880			INDUC BEND	9001899CTD	24	24	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1890			INDUC BEND	9001397CTD	28	28	INDUCTION BEND 90 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1900			INDUC BEND	9001820CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1910			INDUC BEND	9001823CTD	6	10	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1920			INDUC BEND	9001824CTD	12	12	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1930			INDUC BEND	9001825CTD	14	14	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1940			INDUC BEND	9001826CTD	16	18	INDUCTION BEND 45 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1950			INDUC BEND	9001827CTD	20	20	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1960	U04		INDUC BEND	9005326CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1970			INDUC BEND	9005326CTD	24	24	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1980	U04		INDUC BEND	9007006CTD	20	20	INDUCTION BEND 45DEG X 10D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
1990	U04		INDUC BEND	9007021CTD	20	20	INDUCTION BEND 45DEG X 15D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2000	U04		INDUC BEND	9007036CTD	20	20	INDUCTION BEND 90 DEG X 10D RADIUS WT 9.53MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2010			INDUC BEND	9004001CTD	28	28	INDUCTION BEND 45 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2020			INDUC BEND	9001830CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2030			INDUC BEND	9001833CTD	6	10	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2040			INDUC BEND	9001834CTD	12	12	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.14 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2050			INDUC BEND	9001835CTD	14	14	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2060			INDUC BEND	9001836CTD	16	18	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 8.74 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2070			INDUC BEND	9001837CTD	20	20	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2080			INDUC BEND	9005331CTD	24	24	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.13 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2090			INDUC BEND	9002001CTD	28	28	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 11.91 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
2100							***** OLETS ABOVE AND BELOW GROUND*****			
2100										
2110										
2110	U04		OLETS	9006400	0.75	0.75	WELDOLET WT 6.02 MM / SCH 160 LTCS A694 F60	4	4	
2120			OLETS	9006219	1	1.5	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4	4	
2130			OLETS	9002450	0.75	0.75	WELDOLET WT 6.35 MM/SCH 160 WT LTCS A694 F60	6	10	
2140			OLETS	9006407	1	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	10	
2150			OLETS	9006409	3	3	WELDOLET WT 6.35 MM/STD WT LTCS A694 F60	8	10	
2160			OLETS	9006408	4	4	WELDOLET WT 6.35 MM/6.02 MM LTCS A694 F60	10	10	
2170			OLETS	9006817	0.75	0.75	WELDOLET WT 7.14 MM/ SCH 160 WT LTCS A694 F60	12	12	
2180			OLETS	9006413	1	2	WELDOLET WT 7.14 MM/XS WT LTCS A694 F60	12	12	
2190			OLETS	9006412	3	3	WELDOLET WT 7.14 MM/STD WT LTCS A694 F60	12	12	
2200			OLETS	9006411	4	4	WELDOLET WT 7.14 MM/6.02 MM LTCS A694 F60	12	12	
2210			OLETS	9006818	0.75	0.75	WELDOLET WT 7.92 MM/ SCH 160 WT LTCS A694 F60	14	14	
2220			OLETS	9002552	1	2	WELDOLET WT 7.92MM/ XS WT LTCS A694 F60	14	14	
2230			OLETS	9006415	3	3	WELDOLET WT 7.92 MM/STD WT LTCS A694 F60	14	14	
2240			OLETS	9006417	4	4	WELDOLET WT 7.92 MM/6.02 MM LTCS A694 F60	14	14	
2250			OLETS	9002455	0.75	0.75	WELDOLET WT 8.74MM/ SCH 160 LTCS A694 F60	16	18	
2260			OLETS	9006423	1	2	WELDOLET WT 8.74 MM/XS WT LTCS A694 F60	16	18	
2270			OLETS	9006422	3	3	WELDOLET WT 8.74 MM/STD WT LTCS A694 F60	16	18	
2280			OLETS	9006421	4	4	WELDOLET WT 8.74 MM/6.02 MM LTCS A694 F60	16	18	
2290			OLETS	9002008	6	6	WELDOLET WT 8.74 MM/6.35 MM LTCS A694 F60	18	18	
2300			OLETS	9002457	0.75	0.75	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	20	20	
2310			OLETS	9006429	1	2	WELDOLET WT 9.53 MM/XS WT LTCS A694 F60	20	20	
2320			OLETS	9006428	3	3	WELDOLET WT 9.53 MM/STD WT LTCS A694 F60	20	20	
2330			OLETS	9006427	4	4	WELDOLET WT 9.53 MM/6.02 MM LTCS A694 F60	20	20	
2340			OLETS	9006424	6	6	WELDOLET WT 9.53 MM/6.35 MM LTCS A694 F60	20	20	
2350			OLETS	9002076	0.75	0.75	WELDOLET WT 11.13MM/ SCH 160 LTCS A694 F60	24	24	
2360			OLETS	9005119	1	2	WELDOLET WT 11.13 MM/ SCH XS WT LTCS A694 F60	24	24	
2370			OLETS	9005086	3	3	WELDOLET WT 11.13 MM/STD WT LTCS A694 F60	24	24	
2380			OLETS	9006802	4	4	WELDOLET WT 11.13 MM/ 6.02 MM WT LTCS A694 F60	24	24	
2390			OLETS	9005120	6	8	WELDOLET WT 11.13 MM/ 6.35 MM WT LTCS A694 F60	24	24	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 95 °C	12/16/2024	BY OZAC	U04	Page 4 of 6	300K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
2400			OLETS	9002004	4	4	WELDOLET WT 11.91MM/6.02MM LTCS A694 F60	28	28		
2410			OLETS	9002009	6	10	WELDOLET WT 11.91MM/6.35MM LTCS A694 F60	28	28		
2420	U04		OLETS	9008202	1	1	WELDOLET FLAT BOTTOM SCH XS LTCS A694 F60	0	0		
2430											
2430	U04		OLETS	5504450	0.75	0.75	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1	2	2		
2440	U04		OLETS	5504451	0.75	0.75	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2 CL.1	3	3		
2450			OLETS	9006807	0.75	0.75	NIPOFLANGE 300# RF 6.02 MM/SCH 160 WT LTCS A694 F60	4	4		
2460			OLETS	9002643	0.75	2	NIPOFLANGE 300# RF 6.35 MM/XS WT LTCS A694 F60	6	10		
2470			OLETS	9006809	0.75	0.75	NIPOFLANGE 300# RF 7.14 MM/SCH 160 WT LTCS A694 F60	12	12		
2480			OLETS	9006810	0.75	0.75	NIPOFLANGE 300# RF 7.92 MM/SCH 160 WT LTCS A694 F60	14	14		
2490			OLETS	9006811	0.75	0.75	NIPOFLANGE 300# RF 8.74 MM/SCH 160 WT LTCS A694 F60	16	18		
2500			OLETS	9006812	0.75	0.75	NIPOFLANGE 300# RF 9.53 MM/SCH 160 WT LTCS A694 F60	20	20		
2510			OLETS	9006815	0.75	0.75	NIPOFLANGE 300# RF 11.13 MM/SCH 160 WT LTCS A694 F60	24	24		
2520			OLETS	9006814	0.75	0.75	NIPOFLANGE 300# RF 11.91 MM/SCH 160 WT LTCS A694 F60	28	28		
2530	U04		OLETS	5504020	1	1	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	2	2		
2540	U04		OLETS	5504021	1	2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3		
2550			OLETS	9002639	1	2	NIPOFLANGE 300# RF 6.02 MM/XS WT LTCS A694 F60	4	4		
2560			OLETS	9002644	1	2	NIPOFLANGE 300# RF 7.14 MM/XS WT LTCS A694 F60	12	12		
2570			OLETS	9002645	1	2	NIPOFLANGE 300# RF 7.92 MM/XS WT LTCS A694 F60	14	14		
2580			OLETS	9002646	1	2	NIPOFLANGE 300# RF 8.74 MM/XS WT LTCS A694 F60	16	18		
2590			OLETS	9002647	1	2	NIPOFLANGE 300# RF 9.53 MM/XS WT LTCS A694 F60	20	20		
2600			OLETS	9002649	1	2	NIPOFLANGE 300# RF 11.13 MM/XS WT LTCS A694 F60	24	24		
2610			OLETS	9002010	1	2	NIPOFLANGE 300# RF 11.91 MM/XS WT LTCS A694 F60	28	28		
2620	U04			9002070	1	2	NIPOFLANGE 600# RF WT 6.35MM/SCH 160 LTCS A694 F60	6	6		
2630	U04			9003454	1	2	NIPOFLANGE 600# RF WT 8.74 MM/SCH 160 LTCS A694 F60	16	18		
2640		FF5	OLETS	9504524	1	1	NIPOFLANGE FLAT BOTTOM 300# RF SCH XS LTCS A694 F60	0	0		
2650											
2650							***** FLANGES ABOVE & BELOW GROUND*****				
2660											
2660	U04		FLANGE	5556480	0.5	0.75	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 160	0	0		
2670	U04		FLANGE	5556476	1	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
2680	U04		FLANGE	5556473	3	3	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
2690			FLANGE	9001480	4	4	FLG WN 300# RF LTCS A694 F60 WT 6.02MM	0	0		
2700			FLANGE	9001483	6	10	FLG WN 300# RF LTCS A694 F60 WT 6.35MM	0	0		
2710			FLANGE	9001484	12	12	FLG WN 300# RF LTCS A694 F60 WT 7.14MM	0	0		
2720			FLANGE	9001455	14	14	FLG WN 300# RF LTCS A694 F60 WT 7.92MM	0	0		
2730			FLANGE	9001457	8	18	FLG WN 300# RF LTCS A694 F60 WT 8.74MM	0	0		
2740			FLANGE	9001458	20	20	FLG WN 300# RF LTCS A694 F60 WT 9.53MM	0	0		
2750	U04	821	FLANGE	9006232	20	20	FLG WN 300# RF LTCS A694 F60 WT 15.09MM	0	0		
2760	U04	821	FLANGE	9006231	16	16	FLG WN 300# RF LTCS A694 F60 WT 15.88MM	0	0		
2770	U04	821	FLANGE	9003441	10	10	FLG WN 300# RF LTCS A694 F60 WT 21.43MM	0	0		
2780			FLANGE	9001454	10	24	FLG WN 300# RF LTCS A694 F60 WT 11.13MM	0	0		
2790			FLANGE	9002011	28	28	FLG WN 300# RF LTCS A694 F60 WT 11.91MM SERIES A	0	0		
2800	U04		BLIND	5615576	0.5	3	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0		
2810			BLIND	9001450	4	24	FLG BLIND 300# RF LTCS A694 F60	0	0		
2820			BLIND	9001451	28	28	FLG BLIND 300# RF LTCS A694 F60 SERIES A	0	0		
2830	U04		TEE RED	5599703	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	1.5		
2840	U04	FS1	FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
2850	U04	FS1	FLANGE	5557036	1	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
2860	U04	FS1	FLANGE	9001723	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0		
2870	U04	FS1	FLANGE	9001471	6	8	FLG WN 600# RF LTCS A694 F60 WT 6.35MM	0	0		
2880	U04	FS1	FLANGE	5555935	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD	0	0		
2890	U04	FS1	FLANGE	9006755	6	10	FLG WN 900# RF LTCS A694 F60 WT 6.35MM	0	0		
2900	U04	FS1	FLANGE	9002847	16	16	FLG WN 900# RF LTCS A694 F60 WT 8.74MM	0	0		
2910		FS1	FLANGE	9003484	16	16	FLG WN 600# RF LTCS A694 F60 WT 8.74 MM	0	0		
2920		FS1	FLANGE	9003485	20	20	FLG WN 600# RF LTCS A694 F60 WT 9.53 MM	0	0		
2930		FS1	FLANGE	9003487	24	24	FLG WN 600# RF LTCS A694 F60 WT 11.13 MM	0	0		
2940	U04	FS1	FLANGE	5557393	2	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0		
2950											
2950	U04		FLG ADAPTER	5851203	0.5	0.75	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0		
2960											
2960							*****ORIFICE FLANGE*****				
2970											
2970	U04		ORIFICE FLG	9001558	6	6	WN ORIF SET 300# RF A694 F60 WT 6.35MM	0	0		
2980											
2980							***** GASKETS *****				
2990											
2990			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0		
3000			GASKET	5674032	28	28	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0		
3010	U04		GASKET	5673103	1	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0		
3020	U04		GASKET	5672251	2	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0		
3030	U04		GASKET	5673117	3	16	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 95 °C	12/16/2024	BY OZAC	U04	Page 5 of 6	300K5E

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
3040										
3040		B1					***** BOLTING *****			
3050										
3050	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
3060	U04	821		5675776	0.5	24	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
3070										
3070							***** BLINDS & SPACERS *****			
3080										
3080			SPEC BLIND	5749581	0.75	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
3090										
3090			PADDLE	5750100	12	24	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
3100			PADDLE	5750028	28	28	PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3110			PADDLE	5750200	12	24	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
3120			PADDLE	5750198	28	28	PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A	0	0	
3130	U04		PADDLE	5750040	12	24	PADDLE BLANK 600# RF LTCS A516-GR70	0	0	
3140	U04		PADDLE	5750251	12	24	PADDLE SPACER 600# RF LTCS A516-GR70	0	0	
3150										
3150	U04		SPEC BLIND	5749587	1	1	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
3160										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156, DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 40 BARG. @ 95DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 95 °C	12/16/2024	BY OZAC	U04	Page 6 of 6	300K5E